Remainder Theorem & Factor Theorem

Algebra Worksheet · Grade 10–12

Name: ________________________________ Date: __________________

Learning Objectives

- Use the Remainder Theorem to evaluate a polynomial function at a given value using synthetic division
- Apply the Factor Theorem to determine whether a given binomial is a factor of a polynomial
- Use the Rational Root Theorem and synthetic division to fully factor and solve polynomial equations

Problems

1. Use the Remainder Theorem and synthetic division to find f(2) for the polynomial below.

$$f(x) = x^3 - 4x^2 + 5x + 3$$

2. Use the Remainder Theorem and synthetic division to find f(2) for the polynomial below. Note that the x-squared term is missing — replace it with a zero coefficient.

$$f(x) = -x^3 + 6x - 7$$

3. Use the Remainder Theorem and synthetic division to find f(-3) for the polynomial below.

$$f(x) = 2x^3 + x^2 - 5x + 4$$

4. Determine whether (x - 1) is a factor of the polynomial below using the Factor Theorem. Show your synthetic division work.

$$f(x) = x^3 + 9x^2 + 23x + 15$$

5. Use the Factor Theorem to verify that (x + 1) is a factor of the polynomial below. Then divide completely and write all factors.

$$f(x) = x^3 + 9x^2 + 23x + 15$$

6. List all possible rational roots using the Rational Root Theorem, then use synthetic division to find all real solutions of the polynomial equation below.

$$x^3 + 2x^2 - 5x - 6 = 0$$

7. Use the Remainder Theorem to evaluate f(-2) for the polynomial below, then verify by direct substitution.

$$f(x) = 3x^4 - 2x^3 + x^2 - 4x + 5$$

8. Use the Rational Root Theorem and synthetic division to completely factor the polynomial below, then state all real zeros.

$$f(x) = x^3 - 6x^2 + 11x - 6$$

9. Determine the value of k so that (x - 2) is a factor of the polynomial below.

$$f(x) = x^3 - kx^2 + 3x - 2$$

10. Use the Rational Root Theorem and repeated synthetic division to find all real solutions of the polynomial equation below. The polynomial has one repeated root.

$$x^4 - 6x^3 + 9x^2 + 4x - 12 = 0$$

Remainder Theorem & Factor Theorem — Answer Key

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Answer Key

1. Answer: f(2) = 5

- Set up synthetic division with divisor 2 and coefficients 1, -4, 5, 3
- Bring down 1. Multiply: 1×2=2; add to -4 → -2
- Multiply: -2×2=-4; add to 5 → 1
- Multiply: 1×2=2; add to 3 → 5
- The remainder is 5, so by the Remainder Theorem f(2) = 5

2. Answer: f(2) = -3

- Rewrite as f(x) = -x^3 + 0x^2 + 6x - 7; coefficients are -1, 0, 6, -7
- Set up synthetic division with divisor 2
- Bring down -1. Multiply: -1×2=-2; add to 0 → -2
- Multiply: -2×2=-4; add to 6 → 2
- Multiply: 2×2=4; add to -7 → -3
- The remainder is -3, so f(2) = -3

3. Answer: f(-3) = -38

- Coefficients: 2, 1, -5, 4; divisor: -3
- Bring down 2. Multiply: 2×(-3)=-6; add to 1 → -5
- Multiply: -5×(-3)=15; add to -5 → 10
- Multiply: 10×(-3)=-30; add to 4 → -26
- Wait — recalculate: 2, 1×(-3): bring 2, -6+1=-5, 15+(-5)=10, -30+4=-26
- The remainder is -26, so f(-3) = -26

4. Answer: f(1) = 48, so (x - 1) is NOT a factor

- By the Factor Theorem, (x-1) is a factor only if f(1) = 0
- Coefficients: 1, 9, 23, 15; divisor: 1
- Bring down 1. 1+9=10; 10+23=33; 33+15=48
- Remainder = 48 ≠ 0, so (x-1) is NOT a factor

5. Answer: f(-1) = 0; factors are (x+1)(x+3)(x+5)

- Test x = -1: coefficients 1, 9, 23, 15; divisor -1
- Bring down 1. 1×(-1)=-1; -1+9=8; 8×(-1)=-8; -8+23=15; 15×(-1)=-15; -15+15=0
- Remainder = 0, confirming (x+1) is a factor
- Quotient is x^2 + 8x + 15
- Factor x^2 + 8x + 15 = (x+3)(x+5)
- Full factored form: (x+1)(x+3)(x+5)

6. Answer: $x = -1, x = 2, x = -3$

- Possible rational roots: factors of 6 over factors of 1 → ±1, ±2, ±3, ±6
- Test x = 2: 1, 2, -5, -6 with divisor 2 → 1, 4, 3, 0; remainder 0 ✓
- Quotient: x^2 + 4x + 3 = (x+1)(x+3)
- Solutions: x = 2, x = -1, x = -3

7. Answer: $f(-2) = 85$

- Coefficients: 3, -2, 1, -4, 5; divisor: -2
- Bring down 3. 3×(-2)=-6; -6+(-2)=-8; -8×(-2)=16; 16+1=17; 17×(-2)=-34; -34+(-4)=-38; -38×(-2)=76; 76+5=81
- Recalculate carefully: 3 | -2 | 1 | -4 | 5
- 3; 3×(-2)=-6, -2+(-6)=-8; -8×(-2)=16, 1+16=17; 17×(-2)=-34, -4+(-34)=-38; -38×(-2)=76, 5+76=81
- Remainder = 81, so f(-2) = 81
- Verify: 3(16) - 2(-8) + 1(4) - 4(-2) + 5 = 48+16+4+8+5 = 81 ✓

8. Answer: Zeros: $x = 1, x = 2, x = 3$; $f(x) = (x-1)(x-2)(x-3)$

- Possible rational roots: ±1, ±2, ±3, ±6
- Test x=1: 1, -6, 11, -6 → 1, -5, 6, 0 ✓; quotient x^2 - 5x + 6
- Factor x^2 - 5x + 6 = (x-2)(x-3)
- Zeros: x = 1, 2, 3

9. Answer: $k = 3$

- By the Factor Theorem, f(2) must equal 0
- f(2) = (2)^3 - k(2)^2 + 3(2) - 2 = 0
- 8 - 4k + 6 - 2 = 0
- 12 - 4k = 0
- 4k = 12, so k = 3

10. Answer: $x = -1, x = 2$ (multiplicity 2), $x = 3$

- Possible rational roots: ±1, ±2, ±3, ±4, ±6, ±12
- Test x=2: 1, -6, 9, 4, -12 → 1, -4, 1, 6, 0 ✓; quotient x^3 - 4x^2 + x + 6
- Test x=2 again on x^3 - 4x^2 + x + 6: 1, -4, 1, 6 → 1, -2, -3, 0 ✓; quotient x^2 - 2x - 3
- Factor x^2 - 2x - 3 = (x-3)(x+1)
- All solutions: x = 2 (multiplicity 2), x = 3, x = -1